

# CT-EBM-SP - Corpus of Clinical Trials for Evidence-Based-Medicine in Spanish (version 3)

## Project documentation

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Companion file to article:

### Transformer-based relation extraction and concept normalization using an annotated clinical trials corpus

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## Related Annotation Work

| Corpus                                     | Text type and size                                                                                             | Normalizations (count)                                                             |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| BioCreative <sup>1</sup>                   | 6218 journal abstracts.                                                                                        | Normalization to yeast, fly and mouse genes identifiers.                           |
| CANTEMIST <sup>2</sup>                     | 1301 anonymized oncology clinical cases.                                                                       | 16030 entities normalized to the ICD-O-3 tumour classification (850 unique codes). |
| DisTEMIST <sup>3</sup>                     | 1000 anonymized clinical cases extracted from journal articles.                                                | 10 665 entities normalized to SNOMED-CT (7303 unique codes).                       |
| NCBI corpus <sup>4</sup>                   | 793 abstracts.                                                                                                 | Disease entities (6892) normalized to MeSH and OMIM.                               |
| CRAFT <sup>5</sup>                         | 67 full-text journal articles.                                                                                 | Nearly 100 000 concepts from the Gene Ontology.                                    |
| ShARe (SemEval 2015 Task 14 <sup>6</sup> ) | 531 clinical notes.                                                                                            | Disorder entities (19 111) normalized to the UMLS.                                 |
| MCN corpus <sup>7</sup>                    | 100 discharge summaries.                                                                                       | 10 919 concept mentions normalized to the UMLS.                                    |
| QUAERO <sup>8</sup>                        | 2500 Medline titles, 13 texts from EMEA, and 25 EPO patents in French                                          | 26 409 entity annotations were mapped to 26 281 UMLS CUIs (5797 unique codes).     |
| MANTRA <sup>9</sup>                        | 1450 texts from EMEA, Medline and biomedical patent claims in English, French, German, Spanish and Dutch.      | 5530 annotations (total for all languages) normalized to the UMLS.                 |
| E3C <sup>10</sup>                          | 422 anonymized clinical cases extracted from journal articles in Basque, English, French, Italian and Spanish. | 6475 entities (total for all languages) normalized to the UMLS.                    |

Acronyms: *CT*: clinical trials; *CUI*: Concept Unique Identifier; *E3C*: European Clinical Case Corpus; *EMEA*: European Medicines Agency; *ICD-O-3*: International Classification of Diseases for Oncology vs 3; *EPO*: European Patent Office; *MCN*: Medical Concept Normalization; *MeSH*: Medical Subject Headings; *NCBI*: National Center for Biotechnology Information; *OMIM*: Online Mendelian Inheritance in Man; *SNOMED-CT*: Systematized Nomenclature of Medicine – Clinical Terms; *UMLS*: Unified Medical Language System.

**Table 1.** Biomedical and CT-related corpora for medical concept normalization.

| Corpus                           | Text type and size                                                  | Annotations (count)                                                                                                                                                    |
|----------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i2b2 <sup>11</sup>               | Patient reports (394 in train + 477 in test).                       | 3 entity types (in total, 27 837 train, 5009 test), and 6 relation types (5264 train; 9069 test), as reported in <sup>12</sup> .                                       |
| ADE <sup>13</sup>                | Medical reports (4272 sentences).                                   | 3 entity types: Drug (5063), Dosage (231) and Adverse-Effect (5776); 2 relation types: Adverse-Effect (6821) and Drug-dosage (279).                                    |
| DDI <sup>14</sup>                | 1025 documents from DrugBank and Medline.                           | 4 entity types (18 502) and 4 types of DDI relationships (5028).                                                                                                       |
| CDR <sup>15</sup>                | 1500 abstracts from Pubmed.                                         | Chemical entities (4409), Disease entities (5818) and Chemical-Disease relations (3116).                                                                               |
| n2c2 2018 ADE <sup>16</sup>      | 505 discharge summaries.                                            | 9 entity types (83 869) and 8 relation types (59 810).                                                                                                                 |
| Chia <sup>17</sup>               | 1000 texts from ClinicalTrials.gov (12 409 eligibility criteria).   | 15 entity types (41 487) and 12 different relationships (25 017).                                                                                                      |
| Tseo et al. (2020) <sup>18</sup> | 3314 trials from ClinicalTrials.gov                                 | 10 clinical entity types, 5 attributes and value limits (121 221); negation relation (5054).                                                                           |
| Tseo et al. (2020) <sup>19</sup> | 470 study protocols from a private database.                        | 15 clinical entity types and 7 relation types.                                                                                                                         |
| Nye et al. (2021) <sup>20</sup>  | 1932 journal abstracts describing CTs.                              | Entities describing interventions and outcomes (13895), and relations between them (5054).                                                                             |
| BioRED <sup>21</sup>             | 600 PubMed abstracts.                                               | 6 entity types (20 419), 8 relationships (6503) and relation pairs with novelty findings (4532).                                                                       |
| SNPPhenA <sup>22</sup>           | 360 abstracts from PubMed.                                          | SNPs (868), phenotypes (590), SNP-Phenotype association candidates (1300), neutral (308), negative (120) and positive candidates (872); negation and modality markers. |
| DGO <sup>23</sup>                | 1622 oncology exams and breast cancer notes in Portuguese.          | 10 entity types (146 769) and 3 relation types (111 716).                                                                                                              |
| LEAF <sup>24</sup>               | 1006 CT descriptions from ClinicalTrials.gov (105 816 annotations). | 50 entity types (41 487) and 51 relation types (25 017).                                                                                                               |
| RCT-ART <sup>25</sup>            | 558 sentences from 6 disease areas.                                 | 3 entity types (3541) and 3 relationships (3182).                                                                                                                      |

**Table 2.** Biomedical and CT-related corpora annotated with relationships.

Acronyms: *ADE*: Adverse Drug Effects; *CDR*: Chemical-Disease Relations; *CTs*: Clinical Trials; *DDI*: Drug-Drug Interactions; *SNPs*: Single Nucleotide Polymorphisms.

## Annotation Scheme

|                   |                                                                                                    |                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| UMLS              | ANAT: Anatomical structure, body substance or cell.                                                | <i>músculo</i> ('muscle')                                       |
|                   | CHEM: Chemical or pharmacological substance.                                                       | <i>antibiótico</i> ('antibiotic')                               |
|                   | DEVI: Medical device.                                                                              | <i>sonda</i> ('probe')                                          |
|                   | DISO: Pathology, disorder, neoplasms, injuries or signs and symptoms.                              | <i>cancer, fiebre</i> ('fever')                                 |
|                   | GENE: Nucleotide sequence, gene or genome.                                                         | <i>HER2, BRAF</i>                                               |
|                   | LIVB: Human; professional or population group; animal, bacterium, fungus; plant and virus.         | <i>paciente</i> ('patient'),<br><i>S. aureus</i>                |
|                   | PHYS: Biologic, cell, genetic or physiologic function.                                             | <i>digestión</i> ('digestion')                                  |
|                   | PROC: Diagnostic, therapeutic or laboratory procedure; health care activity; research activity.    | <i>hemograma</i> ('hemogram'),<br><i>bypass</i>                 |
| Intervention      | Contraindicated: Any procedure or drug that is a contraindication. [Attribute]                     | <i>Contraindicación a AAS</i><br>(('ASA contraindication'))     |
|                   | Dose_or_Strength: Drug dose or strength.                                                           | <i>2mg, 2%</i>                                                  |
|                   | Form: Dosage form.                                                                                 | <i>pastilla</i> ('pill')                                        |
|                   | Route_or_Mode: Route or mode of administration.                                                    | <i>per os</i> ('oral')                                          |
| Temporal          | Age. [entity or attribute]                                                                         | <i>18 años</i> ('18 y.o.')                                      |
|                   | Date: Specific or relative date.                                                                   | <i>2020, hoy</i> ('today')                                      |
|                   | Duration_or_Interval: length of time that an event lasts.                                          | <i>6 meses</i> ('6 months')                                     |
|                   | Frequency: How often an event occurs.                                                              | <i>semanal</i> ('weekly')                                       |
|                   | Time: Hour or part of the day.                                                                     | <i>15:00</i>                                                    |
| Assertion         | Neg_cue: Negation cue.                                                                             | <i>no</i>                                                       |
|                   | Negated: Negated concept or event. [Attribute]                                                     | <i>sin fiebre</i> ('no fever')                                  |
|                   | Spec_cue: Speculation cue.                                                                         | <i>posible</i> ('possible')                                     |
|                   | Speculated: Speculated concept or event. [Attribute]                                               | <i>probable diabetes</i>                                        |
| Event temporality | Family_History_of: Record of medical conditions or procedures in a patient's family. [Attribute]   | <i>Antecedentes familiares: DB</i><br>(('Family history: DB'))  |
|                   | Future: Any expected event or a pending procedure. [Attribute]                                     | <i>embarazo planificado</i><br>(('planned pregnancy'))          |
|                   | History_of: Record of medical conditions or procedures. [Attribute]                                | <i>Antecedentes de ictus</i><br>(('history of stroke'))         |
|                   | Hypothetical: Any event that is imagined, but it is not truly happening at the moment. [Attribute] | <i>si es mujer, debe...</i><br>(('If female, subject must...')) |
|                   |                                                                                                    |                                                                 |
| Experiencer       | Family_member: A patient's relative. [Attribute]                                                   | <i>padre</i> ('father')                                         |
|                   | Other: Any person that is not the patient/participant nor a relative. [Attribute]                  | <i>representante legal</i><br>(('legal representative'))        |
|                   | Patient: ill person or participant in a trial. [Attribute]                                         | <i>pacientes pediátricos</i><br>(('pediatric patients'))        |
| Other             | Concept: idea, evaluation scale, procedure approach or temporal concept not in TimeML.             | <i>esperanza de vida</i><br>(('life expectancy'))               |
|                   | Food_or_Drink                                                                                      | <i>soja</i> ('soy'), <i>alcohol</i>                             |
|                   | Observation_or_Finding: Clinical findings or conditions that are not pathological states.          | <i>normotenso</i><br>(('normal pressure'))                      |
|                   | Quantifier_or_Qualifier: modifiers/adjectives needed to understand the text.                       | <i>grave</i> ('severe')                                         |
|                   | Result_or_Value: Outcome of a lab or a procedure.                                                  | <i>&gt; 140 kg</i>                                              |

**Table 3.** Annotated entities and attributes with examples.

|                                 | Relation type               | Examples                                                       |
|---------------------------------|-----------------------------|----------------------------------------------------------------|
| Assertion                       | Negation                    | <i>no</i> → <i>fiebre</i><br>(‘no fever’)                      |
|                                 | Speculation                 | <i>posible</i> → <i>diabetes</i><br>(‘possible diabetes’)      |
| Intervention-related / temporal | Has_Dose_or_Strength        | <i>aspirina</i> → <i>500 mg</i>                                |
|                                 | Has_Form                    | <i>aspirina</i> → <i>comprimido</i> (‘tablet’)                 |
|                                 | Has_Route_or_Mode           | <i>aspirina</i> → <i>oral</i>                                  |
|                                 | Has_Duration_or_Interval    | <i>aspirina</i> → <i>un mes</i> (‘one month’)                  |
|                                 | Has_Frequency               | <i>aspirina</i> → <i>diaria</i> (‘daily’)                      |
|                                 | Has_Age                     | <i>paciente</i> (‘patient’) → <i>50 años</i> (‘50 y.o.’)       |
|                                 | After                       | <i>sutura</i> (‘suture’) → <i>operación</i> (‘surgery’)        |
|                                 | Before                      | <i>anestesia</i> (‘anesthesia’) → <i>operación</i> (‘surgery’) |
|                                 | Overlap                     | <i>operación</i> (‘surgery’) → <i>2009</i>                     |
|                                 | Combined_with               | <i>dolutegravir</i> → <i>lamivudina</i> (‘lamivudine’)         |
|                                 | Used_for                    | <i>aspirina</i> → <i>tratamiento</i> (‘treatment’)             |
| Other                           | Causes                      | <i>VIH</i> (‘HIV’) → <i>SIDA</i> (‘AIDS’)                      |
|                                 | Experiences                 | <i>paciente</i> (‘patient’) → <i>diabetes</i>                  |
|                                 | Has_Quantifier_or_Qualifier | <i>asma</i> (‘asthma’) → <i>grave</i> (‘severe’)               |
|                                 | Has_Result_or_Value         | <i>IMC</i> (‘BMI’) → <i>&gt; 25</i>                            |
|                                 | Location_of                 | <i>dolor</i> (‘pain’) → <i>oído</i> (‘ear’)                    |

**Table 4.** Annotated relations with examples.

## Annotation Results

We annotated 87037 entities, including 14518 nested entities (16.68%) and 557 (0.64%) discontinuous entities; 16597 attributes and 68206 relationships. 81.75% of entities were normalized. Tables 5 and 6 report the count of entities, attributes and relations. Table 7 shows the count of normalized entities, including the unclear normalization (i.e., entities marked with the '?' character) and the complex normalization (i.e., entities mapped to two or more UMLS CUIs). Figure 1 also shows a chord diagram with the 20 most frequent relations between entities (we used the normalized CUIs to count the relations, and showed the English preferred term).

|                               | Abstracts | EudraCT | Total          |
|-------------------------------|-----------|---------|----------------|
| ANAT                          | 2780      | 4204    | 6984           |
| CHEM                          | 4391      | 4992    | 9383           |
| DEVI                          | 451       | 193     | 644            |
| DISO                          | 4325      | 8894    | 13219          |
| GENE                          | 16        | 75      | 91             |
| LIVB                          | 3432      | 4951    | 8383           |
| PHYS                          | 887       | 1480    | 2367           |
| PROC                          | 9276      | 10120   | 19396          |
| Contraindicated               | 4         | 263     | 267            |
| Dose_or_Strength              | 875       | 223     | 1098           |
| Form                          | 167       | 70      | 237            |
| Route_or_Mode                 | 548       | 444     | 992            |
| Age                           | 528       | 1051    | 1579           |
| Date                          | 606       | 1301    | 1907           |
| Duration_or_Interval          | 1421      | 1543    | 2964           |
| Frequency                     | 300       | 126     | 426            |
| Time                          | 343       | 72      | 415            |
| Concept                       | 2298      | 1073    | 3371           |
| Food_or_Drink                 | 146       | 44      | 190            |
| Observation                   | 1724      | 2174    | 3898           |
| Quantifier_or_Qualifier       | 1856      | 3108    | 4964           |
| Result_or_Value               | 216       | 1409    | 1625           |
| Neg_cue                       | 1195      | 1564    | 2759           |
| Negated                       | 1584      | 2104    | 3688           |
| Spec_cue                      | 351       | 464     | 815            |
| Speculated                    | 509       | 611     | 1120           |
| Family_history_of             | 0         | 16      | 16             |
| Future                        | 10        | 391     | 401            |
| History_of                    | 60        | 2736    | 2796           |
| Hypothetical                  | 9         | 113     | 122            |
| Family_member                 | 30        | 64      | 94             |
| Other                         | 78        | 388     | 466            |
| Patient                       | 3007      | 3950    | 6957           |
| <b>Total entities</b>         | 37817     | 49220   | 87037          |
| <b>Nested entities</b>        | 6153      | 8365    | 14518 (16.68%) |
| <b>Discontinuous entities</b> | 347       | 210     | 557 (0.64%)    |
| <b>Total attributes</b>       | 5606      | 10991   | 16597          |

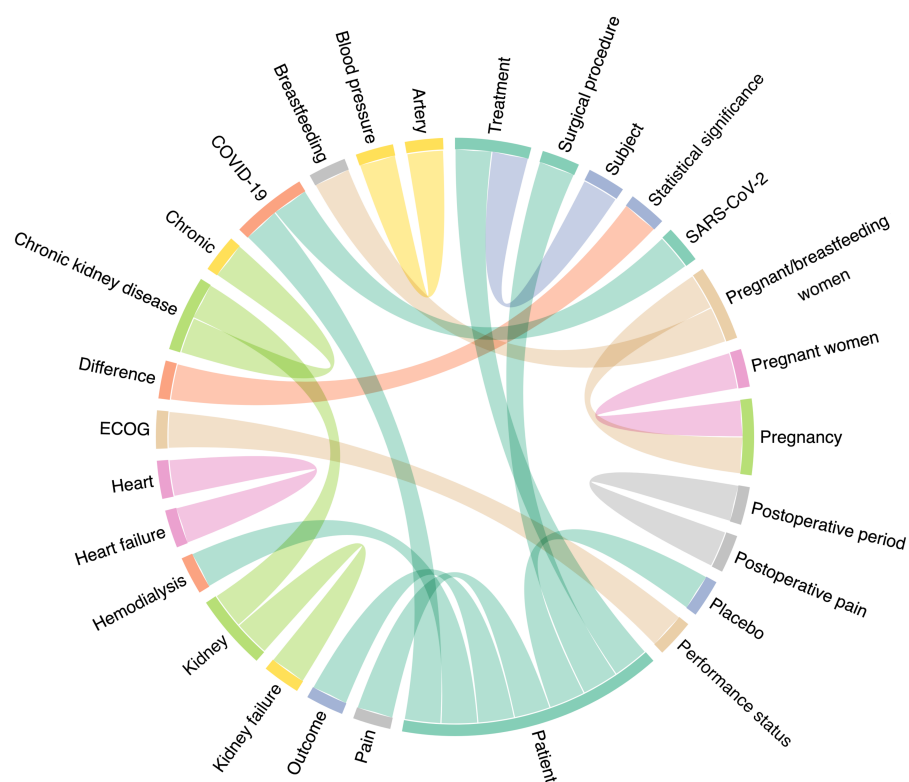
**Table 5.** Distribution of annotated entities and attributes per type.

|                             | Abstracts    | EudraCT      | Total        |
|-----------------------------|--------------|--------------|--------------|
| After                       | 756          | 453          | 1209         |
| Before                      | 3118         | 2960         | 6078         |
| Causes                      | 1638         | 2180         | 3818         |
| Combined_with               | 756          | 672          | 1428         |
| Experiences                 | 7783         | 9002         | 16785        |
| Has_Age                     | 254          | 573          | 827          |
| Has_Dose_or_Strength        | 923          | 294          | 1217         |
| Has_Drug_Form               | 186          | 84           | 270          |
| Has_Duration_or_Interval    | 1149         | 1348         | 2497         |
| Has_Frequency               | 370          | 167          | 537          |
| Has_Quantifier_or_Qualifier | 1935         | 3720         | 5655         |
| Has_Result_or_Value         | 485          | 1640         | 2125         |
| Has_Route_or_Mode           | 663          | 525          | 1188         |
| Location_of                 | 2833         | 4410         | 7243         |
| Negation                    | 1554         | 2049         | 3603         |
| Overlap                     | 3375         | 4498         | 7873         |
| Speculation                 | 488          | 601          | 1089         |
| Used_for                    | 2251         | 2513         | 4764         |
| <b>Total</b>                | <b>30517</b> | <b>37689</b> | <b>68206</b> |

**Table 6.** Distribution of annotated relations per type.

|                         | Abstracts      | EudraCT        | Total          |
|-------------------------|----------------|----------------|----------------|
| Normalized entities     | 30459 (80.54%) | 40714 (82.72%) | 71156 (81.75%) |
| Not-normalized entities | 7358 (19.46%)  | 8506 (17.28%)  | 15881 (18.25%) |
| Unclear normalization   | 1235 (3.27%)   | 1148 (2.33%)   | 2383 (2.74%)   |
| Complex normalization   | 4223 (11.17%)  | 6158 (12.51%)  | 10381 (11.93%) |
| <b>Total entities</b>   | <b>37817</b>   | <b>49220</b>   | <b>87037</b>   |

**Table 7.** Count of normalized entities.



**Figure 1.** Chord diagram of the 20 most frequent relations between entities.



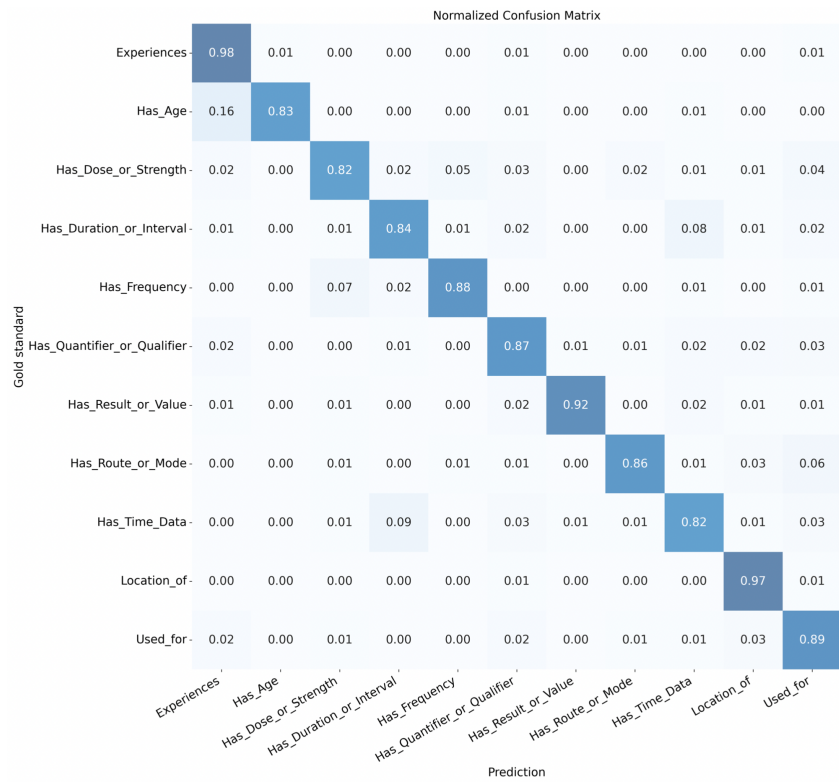
## Relation Extraction Results

|                                              | <b>Precision</b>                | <b>Recall</b>                   | <b>F1</b>                       | <b>Accuracy</b>                 |
|----------------------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
| bert-base-multilingual-cased                 | 0.884<br>( $\pm 0.006$ )        | 0.874<br>( $\pm 0.003$ )        | <b>0.879</b><br>( $\pm 0.005$ ) | <b>0.917</b><br>( $\pm 0.001$ ) |
| bert-base-multilingual-cased vs 2            | 0.880<br>( $\pm 0.006$ )        | <b>0.876</b><br>( $\pm 0.005$ ) | 0.878<br>( $\pm 0.003$ )        | 0.916<br>( $\pm 0.001$ )        |
| dccuchile-bert-base-spanish-wwm-uncased      | 0.868<br>( $\pm 0.009$ )        | 0.857<br>( $\pm 0.006$ )        | 0.862<br>( $\pm 0.006$ )        | 0.907<br>( $\pm 0.003$ )        |
| dccuchile-bert-base-spanish-wwm-uncased vs 2 | 0.877<br>( $\pm 0.008$ )        | 0.854<br>( $\pm 0.005$ )        | 0.864<br>( $\pm 0.006$ )        | 0.908<br>( $\pm 0.003$ )        |
| microsoft-mdeberta-v3-base                   | 0.873<br>( $\pm 0.010$ )        | 0.859<br>( $\pm 0.005$ )        | 0.864<br>( $\pm 0.005$ )        | 0.909<br>( $\pm 0.005$ )        |
| microsoft-mdeberta-v3-base vs 2              | <b>0.886</b><br>( $\pm 0.003$ ) | 0.857<br>( $\pm 0.007$ )        | 0.869<br>( $\pm 0.005$ )        | 0.911<br>( $\pm 0.003$ )        |
| RoBERTa-bsc-bio-ehr-es                       | 0.867<br>( $\pm 0.007$ )        | 0.852<br>( $\pm 0.006$ )        | 0.858<br>( $\pm 0.006$ )        | 0.907<br>( $\pm 0.005$ )        |
| RoBERTa-bsc-bio-ehr-es vs 2                  | 0.875<br>( $\pm 0.009$ )        | 0.858<br>( $\pm 0.004$ )        | 0.865<br>( $\pm 0.006$ )        | 0.910<br>( $\pm 0.004$ )        |

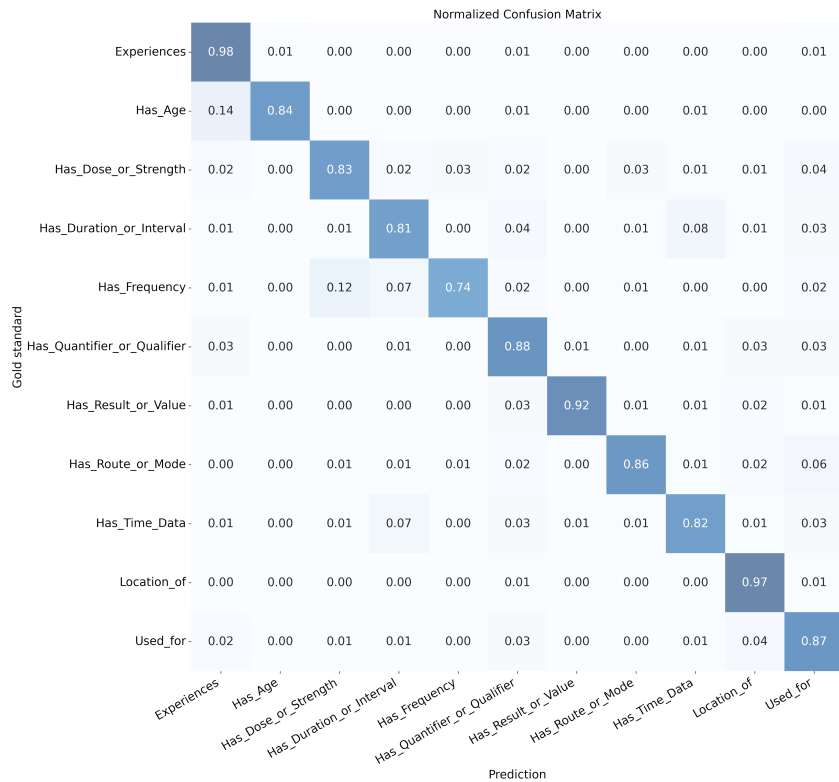
**Table 2.** Results of the relation extraction task (average  $\pm$  standard deviation of 5 experimental rounds).

|                                              |                   |           |                                  |                |                       |
|----------------------------------------------|-------------------|-----------|----------------------------------|----------------|-----------------------|
| <b>One-way analysis of variance</b>          |                   |           |                                  |                |                       |
| P value                                      | < 0.0001          |           |                                  |                |                       |
| P value summary                              | ***               |           |                                  |                |                       |
| Are means signif. different? (P < 0.05)      | Yes               |           |                                  |                |                       |
| Number of groups                             | 8                 |           |                                  |                |                       |
| F                                            | 8.594             |           |                                  |                |                       |
| R square                                     | 0.6528            |           |                                  |                |                       |
| <b>Bartlett's test for equal variances</b>   |                   |           |                                  |                |                       |
| Bartlett's statistic (corrected)             | 1.512             |           |                                  |                |                       |
| P value                                      | 0.9819            |           |                                  |                |                       |
| P value summary                              | ns                |           |                                  |                |                       |
| Do the variances differ signif. (P < 0.05)   | No                |           |                                  |                |                       |
| <b>ANOVA Table</b>                           | <b>SS</b>         | <b>df</b> | <b>MS</b>                        |                |                       |
| Treatment (between columns)                  | 0.001849          | 7         | 0.0002641                        |                |                       |
| Residual (within columns)                    | 0.0009833         | 32        | 3.073e-005                       |                |                       |
| Total                                        | 0.002832          | 39        |                                  |                |                       |
| <b>Bonferroni's Multiple Comparison Test</b> | <b>Mean Diff.</b> | <b>t</b>  | <b>Significant? P &lt; 0.05?</b> | <b>Summary</b> | <b>95% CI of diff</b> |
| mBERT vs mBERT v2                            | 0.001833          | 0.5229    | No                               | ns             | -0.01011 to 0.01378   |
| mBERT vs BSC-BIO                             | 0.01717           | 4.897     | Yes                              | ***            | 0.005220 to 0.02911   |
| mBERT vs BSC-BIO v2                          | 0.01333           | 3.803     | Yes                              | *              | 0.001386 to 0.02528   |
| mBERT vs BETO                                | 0.02083           | 5.942     | Yes                              | ***            | 0.008886 to 0.03278   |
| mBERT vs BETO v2                             | 0.01417           | 4.041     | Yes                              | **             | 0.002220 to 0.02611   |
| mBERT vs mDeBERTa                            | 0.01450           | 4.136     | Yes                              | **             | 0.002553 to 0.02645   |
| mBERT vs mDeBERTa v2                         | 0.009167          | 2.615     | No                               | ns             | -0.002780 to 0.02111  |
| mBERT v2 vs BSC-BIO                          | 0.01533           | 4.374     | Yes                              | **             | 0.003386 to 0.02728   |
| mBERT v2 vs BSC-BIO v2                       | 0.0115            | 3.280     | No                               | ns             | -0.0004470 to 0.02345 |
| mBERT v2 vs BETO                             | 0.01900           | 5.419     | Yes                              | ***            | 0.007053 to 0.03095   |
| mBERT v2 vs BETO v2                          | 0.01233           | 3.518     | Yes                              | *              | 0.0003863 to 0.02428  |
| mBERT v2 vs mDeBERTa                         | 0.01267           | 3.613     | Yes                              | *              | 0.0007195 to 0.02461  |
| mBERT v2 vs mDeBERTa v2                      | 0.007333          | 2.092     | No                               | ns             | -0.004614 to 0.01928  |
| BSC-BIO vs BSC-BIO v2                        | -0.003833         | 1.093     | No                               | ns             | -0.01578 to 0.008114  |
| BSC-BIO vs BETO                              | 0.003667          | 1.046     | No                               | ns             | -0.008280 to 0.01561  |
| BSC-BIO vs BETO v2                           | -0.003000         | 0.8557    | No                               | ns             | -0.01495 to 0.008947  |
| BSC-BIO vs mDeBERTa                          | -0.002667         | 0.7606    | No                               | ns             | -0.01461 to 0.009280  |
| BSC-BIO vs mDeBERTa v2                       | -0.008000         | 2.282     | No                               | ns             | -0.01995 to 0.003947  |
| BSC-BIO v2 vs BETO                           | 0.007500          | 2.139     | No                               | ns             | -0.004447 to 0.01945  |
| BSC-BIO v2 vs BETO v2                        | 0.0008334         | 0.2377    | No                               | ns             | -0.01111 to 0.01278   |
| BSC-BIO v2 vs mDeBERTa                       | 0.001167          | 0.3327    | No                               | ns             | -0.01078 to 0.01311   |
| BSC-BIO v2 vs mDeBERTa v2                    | -0.004167         | 1.188     | No                               | ns             | -0.01611 to 0.007780  |
| BETO vs BETO v2                              | -0.006667         | 1.902     | No                               | ns             | -0.01861 to 0.005280  |
| BETO vs mDeBERTa                             | -0.006333         | 1.806     | No                               | ns             | -0.01828 to 0.005614  |
| BETO vs mDeBERTa v2                          | -0.01167          | 3.328     | No                               | ns             | -0.02361 to 0.0002804 |
| BETO v2 vs mDeBERTa                          | 0.0003332         | 0.09504   | No                               | ns             | -0.01161 to 0.01228   |
| BETO v2 vs mDeBERTa v2                       | -0.005000         | 1.426     | No                               | ns             | -0.01695 to 0.006947  |
| mDeBERTa vs mDeBERTa v2                      | -0.005333         | 1.521     | No                               | ns             | -0.01728 to 0.006614  |

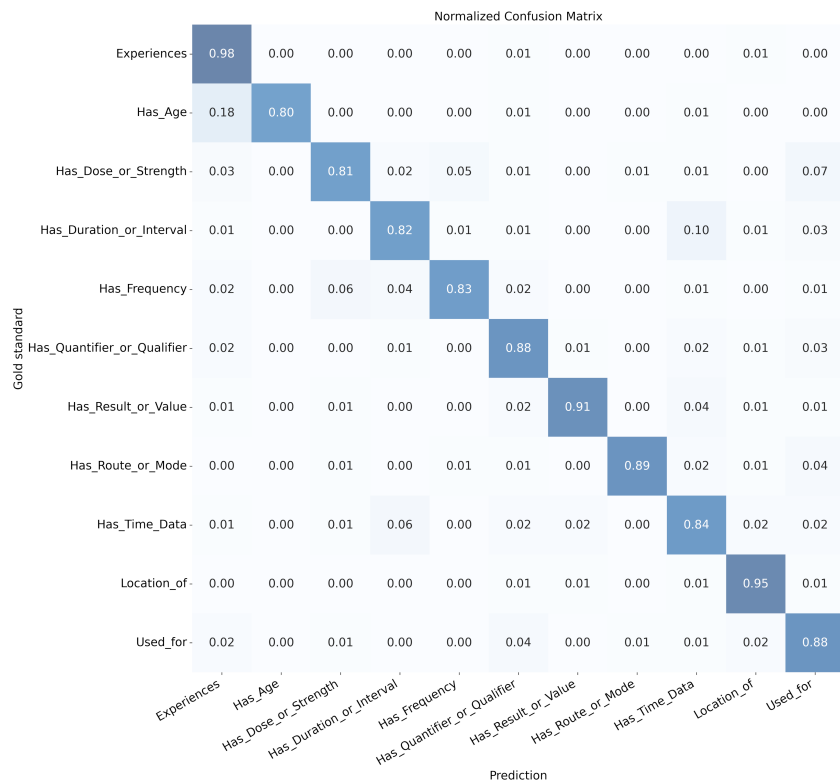
**Table 7.** One-way analysis of variance and Bonferroni's multiple comparison test between models for the F1 measure of the relation extraction task.



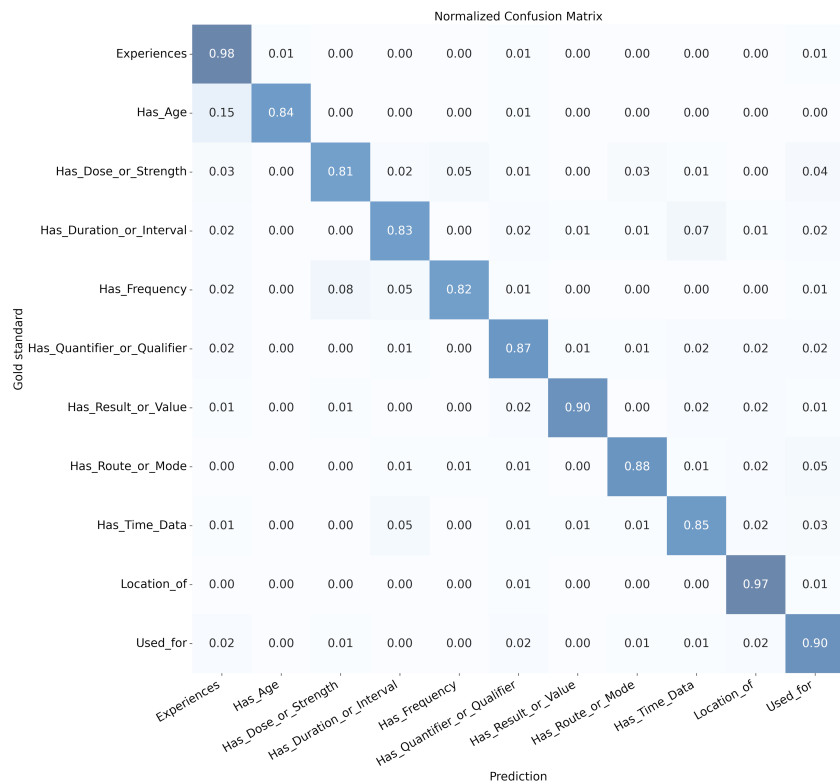
**Figure 2.** Confusion matrix of the bert-base-multilingual-cased vs 2 model.



**Figure 3.** Confusion matrix of the dccuchile-bert-base-spanish-wwm-uncased vs 2 model.



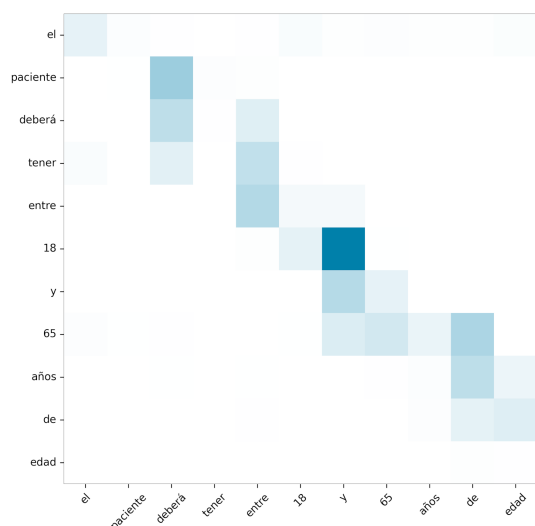
**Figure 4.** Confusion matrix of the microsoft-mdeberta-v3-base vs 2 model.



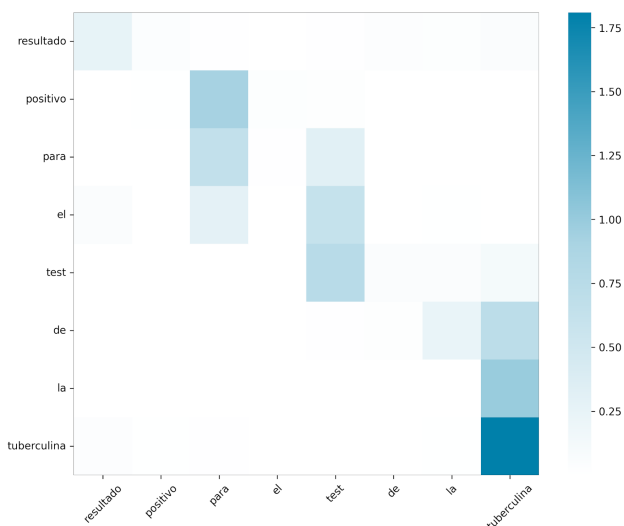
**Figure 5.** Confusion matrix of the RoBERTa-bsc-bio-ehr-es vs 2 model.

## Explainability

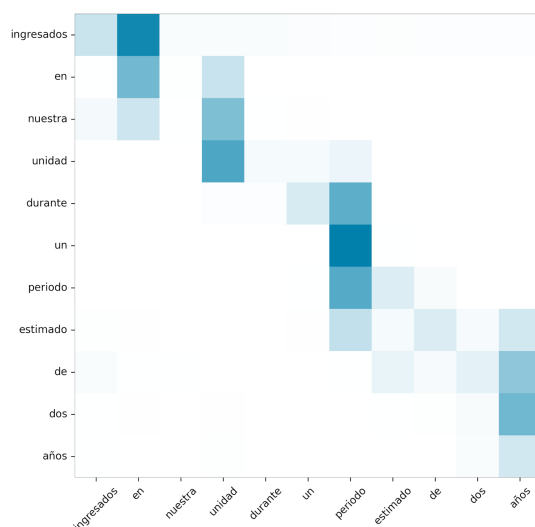
We applied an explainable artificial intelligence (XAI) method to visualize the `bert-base-multilingual-cased` model's attention weights for specific input tokens (Figures S4-S7, Supplementary material). This analysis tends to show that Transformer models focus on key words in the medical entities from the sentence context to predict a relationship.



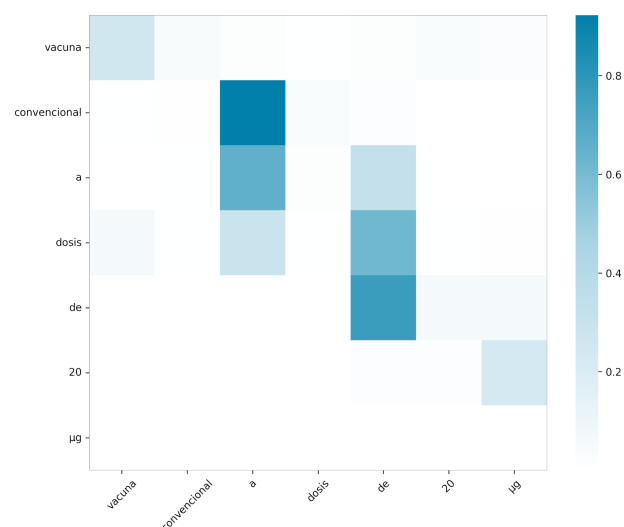
**Figure 6.** Attention weights predicting relation Has\_Age in sentence 'The patient must be between the ages of 18 and 65'.



**Figure 7.** Attention weights predicting relation Has\_Result\_or\_Value in sentence 'Positive result in the tuberculin test'.



**Figure 8.** Attention weights predicting relation Has\_Duration in sentence 'Admitted to our unit for an estimated period of two years'.



**Figure 9.** Attention weights predicting relation Has\_Dose\_or\_Strength in sentence 'Conventional vaccine at a dose of 20 µg'.

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